ALPHA MICROSYSTEMS
RIGHT. FROM THE START

AlphaNET®
User's Guide

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CHAPTER 1

INTRODUCTION AND OVERVIEW

In this chapter, we discuss:

- What is a Network?
- What is AlphaNET?
- What Can AlphaNET Do?
- Graphics Conventions We'll be Using

1.1 WHAT IS A NETWORK?

A network is a way of linking together computers so messages, data, and files can be transferred from one computer to another. A network allows computers to "talk" to one another, without the trouble of using transfer media such as magnetic tape or floppy diskettes.

A network consists of the hardware necessary to link the computers together, plus the software that transfers the data and keeps track of the computers that are part of the network (knowing which ones are "on" and "off" and which ones are transmitting and receiving data).

The network software consists of a number of "layers," which are "between" the physical hardware and the operating system. These layers of software make it possible for a computer to be connected to more than one network at a time—in fact, to more than one kind of network at a time.

1.2 WHAT IS ALPHANET?

Alpha Micro has developed AlphaNET as a method of transferring files, exchanging information, and sharing resources between different Alpha Micro computer systems.

The AlphaNET software is used in conjunction with a choice of several interconnection methods, including asynchronous serial links (SerialNET), Alpha Micro's video technology hardware (VideoNET), and Ethernet.

Each computer system becomes a *node*—one of the connecting links on the network when it is connected to the AlphaNET system. Nodes are arranged in *groups*, which are user-defined sub-sections of a network. Group definitions give you the flexibility to organize and structure your network to meet your communication needs.

Multiple networks can be linked together using *gateways* which transparently connect networks so they can be treated as a single large network.

AlphaNET is a "layered" software product which allows it to take advantage of new interconnection methods, without requiring changes to the way you use AlphaNET.

1.3 WHAT CAN ALPHANET DO?

You can use AlphaNET to:

- Transfer files to and from other computers
- Send messages to users on other computers
- Exchange information by electronic mail
- Get a directory of accounts on other computers
- Send files to printers on other computers for printing. In this way, different computers can share printers.
- Connect your terminal to a remote computer system, making it seem as if you had a direct connection.
- Get status information on remote computer systems.

AlphaNET is designed to make your use of remote computer systems completely transparent. You only need to specify the system a file or user is on to be able to access it. AlphaNET takes care of all network routing.

If you use your own applications programs over the network, be aware that AMOS releases earlier than 2.1 do not provide file or record locking over the network.

A number of AMOS commands are not permitted to be used over the network because of the potential for harm to another system. For example, you are not allowed to run SYSACT and change the account numbers on another computer! If you try to use the following programs:

DSKANA	DSKPAK	FORCE	MOUNT
SUBMIT	SYSACT		

or any of the certification, video backup, or formatting programs (such as CRT420, VCRSAV, or FMT210), you will receive an error message. Some other system commands also may not work over the network because they would have no logical application over a network.

1.4 GRAPHICS CONVENTIONS

This manual conforms to the other Alpha Micro publications in its use of a standard set of graphics conventions. We hope these graphics simplify our examples and make them easier for you to use. Unless stated otherwise, all examples of commands are assumed to be entered at AMOS command level.

SYMBOL	MEANING
devn:	Device-Name. The "dev" is the three letter physical device code, and the "n" is the logical unit number. Examples of device names are DSK0:, DSK5:, WIN1:, and MTU0:. Usually, device names indicate disk drives, but they can also refer to magnetic tape drives and video cassette recorders.
filespec	File Specification. A file specification identifies a specific file within an account. A complete filespec is made up of the devn:, the filename, the file extension, and the project-programmer number. For example: devn:filename.ext[p,pn] -or- DSK0:SYSTEM.INI[1,4]
[p,pn]	This abbreviation represents an account on a disk you can store files and data in. An actual disk account number looks like this: [100,2] or [1,4]. Disk account specifications are sometimes referred to as "Project-programmer numbers."
{}	Braces are used in some examples to indicate optional elements of a command line. In the example: **DIR{/switch}** the braces tell you "/switch" is not a required portion of the DIR command line.

SYMBOL	MEANING
/	The slash symbol precedes a command line switch or "option request." For example: **DIR/WIDE:3** RETURN This command requests a directory display of the disk account you are currently logged into. The switch (/WIDE:3) indicates you want the display to be three columns wide.
TEXT	Bold text in an example of user/computer communication represents the characters you type.
TEXT	Text like this in an example of user/computer communication represents the characters the computer displays on your terminal screen.
KEY	In our examples, the keycap symbol appears whenever you need to press a certain key on your terminal keyboard. The name of the key you need to press appears inside the keycap symbol, like this: RETURN. If you need to press the TAB key, you would see TAB, or the ESCAPE key, ESCAPE. (Sometimes the ESCAPE key is labeled ESC or ALT MODE.)
CTRL/KEY	This indicates a control sequence you press on the keyboard. The CTRL key is pressed and held down while the indicated key is also pressed.
^	This symbol in front of a capital letter means the letter is a "control character." For example, when you press CTRL/C, it appears on your screen as `^C`. (^C is the control character that cancels most programs and returns you to AMOS command level.)
	This symbol means "halt!" It indicates an important note you should read carefully before going further in the documentation. Usually, text next to this symbol contains instructions for something you MUST or MUST NOT do, so read it carefully.
	This symbol means "hint." It indicates a helpful bit of information, or a "short cut" that could save you time or trouble.
	This symbol means "remember." It indicates something you should keep in mind while you are following a set of instructions.

CHAPTER 2

GETTING STARTED WITH AlphaNET

In this chapter, we discuss:

- Finding out the status of your networks
- Getting on and off the network

Once AlphaNET has been installed on your system, you are ready to put it to use. If you have not yet installed AlphaNET, see the *AlphaNET Installation Guide* for installation instructions.

2.1 FINDING OUT THE STATUS OF THE NETWORKS

The NTSTAT program provides you with a status report telling you which networks are available and which systems are currently active on the networks. To see the status display, enter:

NTSTAT RETURN

A typical status display looks something like this:

```
Systems Available:

Video Network:                    Status: Node is on the network
 (self)    4261609474-  Accounting                    AMOS/32 2.0A(217)-4
 SHIP:     4261609475-  Shipping Dept.                AMOS/32 2.0A(217)-4
 RESEAR:   4261675009-  Research and Development      AMOS/32 1.0D(165)-3

Network  2:                       Status: Node is on the network
 (self)      33619970-  Accounting                    AMOS/32 2.0A(217)-4
 SALES:      33619969-  Sales Department              AMOS/32 2.0A(217)-4
 VP:         33619971-  Vice President's System       AMOS/L 2.0A(217)-4

Network  3:                       Status: Node is on the network
 (self)      50397184-  Accounting                    AMOS/32 2.0A(217)-4
 CEO:        50397185-  CEO's System                  AMOS/32 2.0A(217)-4
 MARKET:     50397186-  Marketing                     AMOS/L 1.3D(165)-3

Gateways Available:

 Distribution Center            Forward to MARKET:
 Government Data Center         Forward to RESEAR:
```

In the display above, your computer is listed first (self). The number, 4261609474-, is your cpuID. "Accounting" is the identifying name assigned to your computer in the SYS:SYSNAM.DAT file. The last display on the line tells you what operating system you are running under.

Two other computers, SHIP: (Shipping) and RESEAR: (Research and Development) are also listed. There may be other computers on that network not shown here because they are currently not connected. See NOLINK, below. As you can see, there are other networks, and even gateway connections, also listed in the display.

Each computer shown on the NTSTAT display has a physical connection to the network, but is not necessarily currently active on the network.

If you have a network with a large number of users or if your computer is part of several networks, the network status information displayed by NTSTAT will not fit on a single display screen. To display the network status one page at a time, you can follow the NTSTAT command with the **/P**switch.

If you want to see a continuous, dynamic display of information about the network, you can specify the /C switch with the NTSTAT program. The display will look something like this:

```
                              AlphaNET Status
                                02:15:53 PM
 Accounting Network                                  Node count:   6

 Status: Node is on the network                 Uptime: 04:00:43

 +-------------------------------------------------------------+
 | Block Writes:              22     Block Reads:             34 |
 | Network Adds:               4     Network Deletes:          5 |
 | CRC Errors:                 0     Protocol Errors:          0 |
 | Lost Packets:               0     Framing Errors:           0 |
 | Transfer of Control:    40668     Transmit Errors:          0 |
 | Last Node Added:       HIRES:     Last Node Deleted:    TECH: |
 +-------------------------------------------------------------+

Group  Node    CPU        Ersatz   Group  Node   CPU       Ersatz
-----------------------------------------------------------------
  2     1   1694629889-   (self)     2     9   1694629897-  RODDEV:
  1     2   1694564354-   COMLB2:    2    10   1694629898-  ENGHBF:
  2     2   1694629890-   FS2000:    2    11   1694629899-  DUNC:
```

When using /C, you can use ^T and ^R to page forward and backward in the list of available systems, and ESC to switch between different networks.

Network Specific Information

If you follow NTSTAT with a number (eg. NTSTAT 101), NTSTAT will only display information relating to that particular network. If you use "NTSTAT number" on a network that doesn't exist you get the message:

```
%Network or gateway not found
```

You can also use the /F switch to display a complete listing of all network parameters. This is primarily of interest when setting up or debugging a network.

If you use NTSTAT/F on an EtherNET network, it now the EtherNET addresses associated with each node.

Example:

```
Networks:

1694629897-   Network 101  Group   2  Node     9   (AM362 @ 26338)
                                      EtherNET address: 0000B1000001
                    Group   2  Node     4, 1694629892- 02CF1FA00079
                    Group   2  Node     2, 1694629890- 02CF1FA00085
                    Group   2  Node     3, 1694629891- 02CF1FA00081
                    Group   3  Node     5, 1694695429- 02CF1F605520
                    Group   2  Node     8, 1694629896- 0000B100000B
                    Group   3  Node     1, 1694695425- 0000C0081E13
                    Group   3  Node     4, 1694695428- 02CF1F605533
                    Group   2  Node     7, 1694629895- 02CF1FA00084
                    Group   2  Node     1, 1694629889- 02CF1FA00038
                    Group   1  Node     2, 1694564354- 02CF1FA00034

         Statistics:

         Block Writes:           2919    Block Reads:         3622
         Network Adds:             18    Network Deletes:        8
         Transfers of Control:      0    Protocol Errors:        0
         CRC Errors:               20    Framing Errors:         2
         Lost Packets:              0
```

Note the EtherNET address following the text "EtherNET address:" is the EtherNET address for the system NTSTAT is being run on.

Note that if you use NTSTAT/F on a network that isn't EtherNET (ie. serial or video) you won't see any change.

There is also a /J switch you can use to see what computer's are connected to your system via a virtual terminal connection. The screen generated by the NTSTAT/J command displays the following information:

- The spawned JOB name.
- The name of the computer for which the spawned job was created.
- The Network, Group, Node, and Socket number associated with the computer which made the connection.

The following is an example of the terminal display screen generated by NTSTAT/J:

```
VIRTUAL TERMINAL SERVICE: CONNECTED JOB PAIRS DISPLAY
=====================================================

Spawned Job   ---------- Connected Remote Job Defined By ---------
   Name       System Name   Network #    Group #   Node #   Socket #

 TSKAAA          TECH:          5           1        2        4
 TSKAAB          SQA:         101           2        5       12
 TSKAAC          DEMO2:         1           1        8        9
```

2.2 TURNING THE NETWORK ON AND OFF

The SET program can only be executed from the OPR: account (DSK0:[1,2]). There are two options allowing you to connect and disconnect your system from a network. The formats are:

SET LINK cpuID- RETURN

and:

SET NOLINK cpuID- RETURN

Where cpuID is the identifying number of your computer on the network you wish to turn on or off. For example:

SET LINK 16842757- RETURN

You must specify the cpuID, or an ersatz name that includes the cpuID.

In AMOS 2.1 and later software releases, the SET LINK and SET NOLINK commands can only be executed from DSK0:[1,2]. If you are running an AM-340 Ethernet controller on an AM-2000 with an AM-182 cache board, or an AM-3000 computer, you must "SET LINK" before turning on the cache via CBEN. Also, once the cache is enabled, you may not turn off the Ethernet network with the SET NOLINK command.

Keep in mind turning off the network disables it for all users on your computer. You will normally want to keep your computer connected to the network at all times; the main use for SET NOLINK is for VideoNET installations which must disconnect from the network before using the video circuitry for backup or restore purposes.

To avoid causing problems for other users on your computer, this operation is best left to a single person, normally that person designated as the "system operator."

2.3 VIDEONET AND VCR BACKUP

One of the limitations of the video cable network is your computer system CANNOT be logically connected to the network if you are going to do a VCR backup or use the CRT610 program to certify a video tape. This is because the network is using the video circuitry, which cannot run both the network and the VCR at the same time.

Therefore, if you wish to use the VCR software, you must disengage the VideoNET network using SET NOLINK:

`SET NOLINK cpuID-` RETURN

You may want to create a command file to perform the steps of your video tape backups or certifications. Make sure the command file contains a SET NOLINK command. You probably want to modify any existing command files also.

Note networks other than VideoNET do not affect your ability to perform video backup functions; only VideoNET must be disconnected.

CHAPTER 3

USING AlphaNET

In this chapter, we will discuss:

- What Can You Do on AlphaNET?
- Using Ersatz Names on AlphaNET
- Protecting Devices from Access

3.1 WHAT CAN YOU DO ON ALPHANET?

The next few chapters will discuss the tasks you can do using AlphaNET. These include:

- Using Ersatz Names
- Using Gateways
- Turning On and Off Network Device Access
- Sending Messages to Users on Other Computer Systems
- Getting Directories of Accounts on Other Systems
- Copying Files to and from Other Systems
- Printing Files Across a Network
- Connecting Your Terminal to Another Computer

3.2 USING ERSATZ DEVICE NAMES

Before we get into the uses of the network, we'll explain what ersatz names are, and how you can use them to make AlphaNET easier to use. An ersatz name is a "substitute" name. Ersatz names allow you to replace a long string of characters with a shorter, easier to remember name. They can save you time and trouble when typing network commands. For example, instead of entering:

```
COPY ACCT.DAT=16842749-:ACCT.DAT RETURN
```

with the properly defined ersatz name, you could enter:

```
COPY ACCT.DAT=PURCH:ACCT.DAT RETURN
```

See your *AlphaNET Installation Guide* for information on defining ersatz names.

3.3 USING GATEWAYS

Network gateways allow multiple AlphaNET networks to be interconnected so all nodes on all networks are available to each other. It does so in a way that makes all nodes appear in the same way. That is, no matter how many different networks are involved, all you need to do is refer to the desired system by its cpuID or ersatz device name.

3.4 PROTECTING DEVICES FROM NETWORK ACCESS

This process should be handled by the System Operator.

If you wish to use the network, but you have files or data you do not wish to allow anybody on another system access to, you can protect them by closing off access to the device they reside on. Say you have a system set up with five logical devices. If you don't want people from other computers to access DSK2:, enter:

```
SET NOACCESS DSK2: RETURN
```

People on your own system will still be able to copy from DSK2:, but nobody can do so across AlphaNET. If you wish to permit access to that device again, enter:

```
SET ACCESS DSK2: RETURN
```

You may also specify an asterisk:

```
SET NOACCESS * RETURN
```

Which means ALL the devices on your system cannot be accessed across AlphaNET. In effect, you would have a system that can use AlphaNET to access other systems, but cannot be accessed by other systems. The DEVTBL command can be used to display the current protection status of your devices.

CHAPTER 4

ACCESSING FILES OVER AlphaNET

In this chapter, we discuss:

- How Files Can be Accessed Across AlphaNET Connections
- How to Get a Directory of Another System
- Copying Files Between Systems
- Printing Over the Network

4.1 ACCESSING FILES OVER ALPHANET

AlphaNET has been designed to make network access to files as much like local access as possible. The only exceptions come when a certain operation either requires direct hardware access (such as formatting disks) or is not allowed for security reasons.

In general, any filespec can include a network cpuID specification, allowing you to specify a remote file with any command. This means you can copy, edit, update, or otherwise use a remote file just as if it was present on your own system.

For security reasons, several conditions must be met, however, before access to remote files is allowed. First, the device on which the remote file resides must have had SET ACCESS done to it to allow remote access. If a device has access allowed, and the remote system is using AMOS 2.0 or later, the file in question must also allow remote access.

Note the default file protection allows read and execute access by remote users, but does not allow writing by remote users. When using the AMOS 2.0 extended directory structure, you can select the protection level you wish to control the level of remote access.

The other important thing to keep in mind when accessing remote files is the lack of multi-user file update protection on remote files. The multi-user protection afforded by LOKSER or AMOS 2.0's integrated record locking are not available on remote files. For this reason, care must be exercised whenever accessing remote files that may be in use by other users.

AlphaNET 2.1 does provide remote multi user file protection, if the remote users are also using AMOS 2.1 or later operating system releases.

Some programs, such as SYSACT and DSKANA, do not allow themselves to be used across the network. This is to prevent remote users from being able to corrupt a system or otherwise bypass system security.

Remember you will want to "pull" files from other systems, rather than trying to "push" them. Attempts to "push" the files will often result in "protection violation" errors when write access is denied by the remote system. This is particularly true when trying to "push" a file in such a way you would need to create a new account. This operation is not allowed and, depending on the revision levels of AMOS involved, can cause file system damage. (It is not possible to prevent all occurrences of this without requiring all networked systems to use AMOS 2.0; a restriction we have chosen not to make.)

Here are some things to keep in mind when accessing files across the network:

1. If you are using an operating system earlier than AMOS 2.0, you can not access files on remote systems running with AMOS 2.0 (or later) operating systems that are configured with extended disk format.

2. With AMOS 2.0 and later software releases, the accounts on the disk drive are not always in sequence, account [1,4] may appear after account [44,44]. This condition may cause problems when accessing files across the network.

   For example:

   If your local system is running AMOSL 1.3D software, you will not be able to access files over the network on systems running AMOS 2.0 (or later) with a traditional file structure, if the accounts on the disk you are accessing are not in sequence. The MFDSEQ program can be used to sequence the account numbers on systems running AMOS 2.0 (or later) software with traditional (non-extended) file structures.

3. You will not be able to access files on a remote system that must be accessed through a gateway, if the files on the remote system are located on a device that is not defined on your local system.

   For example:

   If the files you wish to access on the remote system are located on ESD0: and you do not have an ESD0: device defined on your local device, you will not be able to access the files.

4.1.1 Remote File/Record Locking

When using versions of AMOS earlier than 2.1, there are some restrictions that must be observed when using either LOKSER or AMOS 2.0's integrated locking facility. While these restrictions have been removed in AMOS 2.1 (and later) releases, those using AMOS 2.0A must be aware of the restrictions outlined in the following paragraph.

Files being accessed by users on the same system as the file (local access) are protected from improper access by remote users. However, two remote users, from one or more remote nodes, are not protected from each other. Furthermore, if a remote user opens a file, but does not close it, either because of program error or system crash, the file will remain locked indefinitely. It is then necessary to use LOKUTL to remove the file lock so other users may access the file. For more information about file and record locking see the *AMOS File Locking User's Guide*.

As of AMOS 2.1, file locking for multiple remote users is handled automatically.

4.2 GETTING A DIRECTORY OF ANOTHER SYSTEM

In order to find out what files are available on a remote system, you can use the DIR command across the network, just as you would on a local system. Just include the cpuID (or a valid ersatz name) in the normal file specification:

DIR 12930293-DSK3:[123,6]*.TXT RETURN

or:

DIR LEDGER:DSK0:[1,4] RETURN

The wild card specification **ALL:** is not supported over the network.

4.3 COPYING FILES BETWEEN SYSTEMS

You can use the standard COPY program to copy files across the network. The only difference is you need to specify the cpuID. The format is:

```
COPY source-full filespec
```

For example:

COPY ACCT1.TXT[50,2]=16844425-DSK0:ACTTMP.TXT[100,5] RETURN
COPY WRHREC.DAT=16842756-DSK2:WRHREC.DAT[10,5] RETURN

If you use ersatz device names, you must not follow the colon with a space. The standard AMOS conventions of wildcarding and file specifications apply.

Copying files between systems connected by gateways works exactly the same way.

4.4 PRINTING OVER THE NETWORK

AlphaNET allows you to send a file to another computer system for printing. You may do so using either the Task Manager print spooler, or the memory based print spooler (LPTSPL).

4.4.1 Printing Using the Task Manager Spooler

The format for the task manager printer spooler is:

```
PRNT{/switch} {cpuID-}{printerspec}=
        {/switch}
```

Some examples:

```
PRNT 16842758-LPT1=PAYROL.LST RETURN
PRNT GNLDGR:DIABLO=RESULTS RETURN
PRNT 14932034-LASWRT=MASTER.PAY/C:4 RETURN
```

If you use a printer on another system regularly, you will probably want to set up a DO file to make it easier to send files to the printer.

Because of the procedure PRNT uses to select defaults, the specification can be a little tricky when the system you want to print to has different device names than your own system. If the system you are sending to does not have a device name defined on it with the same device name as you are logged into, you will have to specify a device name that does exist on that remote system (the safest bet is DSK0:, which all systems must have) on the left side of the equal sign.

For example, say you are logged into DSK5:[100,0] and you wanted to send a file to a printer on another system that does not have a DSK5:. If you specify:

```
PRNT 16842758-LASWRT=REPORT.LST RETURN
```

PRNT assumes the output device is the same as the input device (DSK5:, where you are logged into). When it looks for a DSK5: on the other system, it doesn't find it, and stops, giving you an error message:

```
?Cannot INIT DSK5: - device does not exist
```

If you see such a message, try again, specifying a valid device name on the printer's system on the left side of the equal sign. Or you may try logging into DSK0: on your system, and making the print request from there.

As with any AMOS command specification, the PRNT command must be able to access the printer, and the place where the file to be printed is. Therefore, if the disk you want to print from is set to NOACCESS, or the output disk specified on the printer system is set to NOACCESS, you will also see the above error message.

You may optionally place a file called PRINTR.DFT in account [1,4] of your system disk. This file contains a cpuID which the network will use as a default on all PRNT requests. If this file does not exist, or contains a 0, then PRNT will default to your own computer system. This can be useful if the printer you use most often is on another system.

In order to use AlphaWRITE and AlphaCALC to print to a remote printer, you must have version 1.2 or later of these products.

4.4.2 Printing Using the Memory Based Spooler (LPTSPL)

You can set up the old line-printer spooler to allow remote printing, even if your local system is using the task manager spooler.

On each system on which you wish to have access to a remote printer, create an .INI file (of whatever name you like) in DSK0:[1,4] which contains the following:

```
REMOTE = cpuID
NAME = name
FORMS = NORMAL
```

where "cpuID" is the cpuID number (ersatz names are not allowed here) of the system the printer is attached to, and "name" is the name of the printer as defined in the printer initialization file on that remote system. You may also include the DEFAULT command if you wish. All other commands will be ignored.

Be careful when choosing a name for the printer in the printer initialization file. Make sure it is a **unique** name compared to all the other network-accessible printers or any printers defined on your local system.

Once you have created the initialization file, you will need to add an LPTINI statement to your system initialization command file to define the printer each time you boot the system. Unlike when setting up a local printer, you do not need a separate job to execute the LPTINI command. Instead, simply execute the LPTINI command directly in the system initialization command file. The LPTINI command will define the printer and exit, thus continuing with the remainder of the initialization file.

The LPTSPL spooler has two restrictions when used with remote printers: you cannot display a list of a remote printer's queue, and you can not use PRINT across a network using an ERSATZ name in place of the CPU I.D. number.

Make sure the remote printers you define using the LPTINI command are really printers using the old memory based spooler.

CHAPTER 5

CONNECTING TO OTHER COMPUTERS

In this chapter, we discuss:

- How to Connect Your Terminal to a Remote System
- Virtual Terminals
- Virtual Terminal Drivers
- Finding Remote System Connections

5.1 CONNECTING TO A REMOTE SYSTEM

AlphaNET supports a *Virtual Terminal* capability which allows your terminal to appear to be directly connected to a remote system. This lets you do just about any operation you could perform locally.

To connect your terminal to a remote system, use the CONECT program. The command syntax is:

```
CONECT cpuID- {memory}
```

Specify the cpuID of the system you wish to connect to (or its ersatz name) and an optional amount of memory to allocate on the remote system. You can enter the memory amount in number of bytes (i.e., 65535), or with a "K" (kilobytes) or an "M" (megabytes). If no memory size is specified, a default of 64K will be used. Typing CONECT with no arguments will display help information.

If the remote system you are attempting to connect to has a system password, you will be prompted for that password.

If your system makes use of user names, CONECT will attempt to log you in on the remote system under your current user name. If no such user name exists on that remote system, you will need to log yourself in.

Once a connection has been established, you can use it as you would any other terminal connection.

With the virtual terminal capability, you can enter special commands by using a combination of [|] (broken vertical line character), and other keys, much as you would use a [CTRL] or [ESC] sequence.

When you are done using the remote connection, you can disconnect by entering the key sequence: Press [|]. Release then press [D].

If you should need to enter a "|" character, enter "||" (this way, it is not confused as a command key sequence).

Because you can use the CONECT program multiple times to chain through a series of systems, you can press [|] then release and press [P] to display your current connection path.

You can only connect to a system running VTSER. To find out what systems are available for virtual terminal connection, use the FNDVT program described below. For information on VTSER set-up, see the *AlphaNET Installation Guide*.

Special Features

If you are running AlphaNET 2.2, using an AMOS 2.2 (or later) operating system, and using Ethernet as your network data transport device, there are some additional features built into the terminal connection program CONECT. These features include an improvement in the virtual terminal performance and the availability of the command line editor.

To use these new features under CONECT, you will need to add a special TRMDEF statement. You must also create a special communication "zone," which requires one additional statement in your system initialization command file. See the *AlphaNET Installation Guide* for information on adding the special TRMDEF statement and communication zone.

If for some reason there is a system crash on the remote system you are connected to, there will be a delay of approximately 60 seconds, after which you will be disconnected and returned to your local system.

5.1.1 Virtual Terminal Performance

The performance of the virtual terminal connection is directly related to the performance of the underlying network connection. The faster the network, the better the performance. You will also see an effect based on the number of simultaneous virtual terminal connections.

The usual effect of a slow network connection will be a delay in echoing individual characters as they are entered from the keyboard. Terminal output performance is usually quite good even over a very slow network connection.

SerialNET at 19200 baud or faster gives good virtual terminal performance. VideoNET is slow, but usable.

The best possible virtual terminal performance can be achieved by users who meet the following criteria:

- Running AMOS 2.2 or later operating systems.
- Running AlphaNET 2.2 or later.
- Using Ethernet as your data transport mechanism

Users who fall into the above category will enjoy a data transfer rate that is significantly faster than the performance found in earlier AlphaNET releases. Your computer will also be able to support a larger number of virtual terminal connections with much greater efficiency. You will also be able to use the command line editor for remote connections.

5.1.2 Terminal Drivers for Virtual Terminals

The virtual terminal facility does not do terminal emulation or translation, but instead establishes a virtual circuit between the remote system and your terminal. This eliminates a source of problems with terminal and function key compatibility.

Line editor is not supported, when you are attached as virtual terminal on a remote system, unless you are using Ethernet, AlphaNET 2.2 (or later), and AMOS 2.2 (or later).

In order to establish a virtual terminal connection, however, the terminal driver in use by your physical terminal must be available and installed on the remote system. If it isn't, an error message is displayed and the connection refused.

You can ensure a terminal driver is available by defining a dummy terminal using it (for pre-2.0A AMOS systems) or by using the TDVDEF program (under AMOS 2.0A and later). Using the TDVDEF program is described in the *System Commands Reference Manual*.

5.1.3 Virtual Terminals and 8-Bit Characters

If you want to execute an application that uses 8 bit characters over a virtual connection and you are not using an Alpha Micro terminal, it may be necessary to create a special terminal driver for use while connected to a remote system.

The following information is provided to help you determine whether or not you will need to create this special terminal driver for the remote system. This information is directed at software developers and assumes a familiarity with assembly language programming and the AMOS Terminal Service System (Reference AMOS Monitor Calls Manual, Chapter 7).

When in 8 bit mode (T$EXT is set) a function key is expected to consist of a two character sequence, which is a CSI character (233 octal) followed by the function key character. Alpha Micro's 8 bit terminals have been programmed to generate this two character sequence.

While connected to a remote system, each key pressed on the physical terminal passes through the terminal driver and the AMOS terminal service input routines on the local system. Then it is passed on to the remote system where it passes through the comparable routines on the remote system.

The CONECT program puts the local terminal in the following modes in an attempt to preserve the input stream as it comes from the terminal:

T$DAT - data mode
T$XLT - let the application translate function keys
T$EXT - 8 bit character support

With these modes set, if you are using an Alpha Micro terminal, the data received from the terminal will be passed on to the remote system exactly as it is received. Consequently, Alpha Micro terminal drivers will function properly on either the local system or the remote system.

On the other hand, if your terminal does not generate the proper two character sequence and consequently your terminal driver performs the necessary conversion, the input stream that is passed to the remote system will not be identical to that received from the physical terminal. In this case the terminal driver on the remote system will need to be modified to look for this altered character sequence.

5.1.4 Restrictions on Virtual Terminals

Certain programs may not be used when connected by virtual terminal. These are programs which perform JLOCK operations, locking out all other jobs. Because the VTSER job must be able to run in order to service the virtual terminal connection, executing a program which performs a JLOCK will result in the virtual terminal connection being terminated.

The following programs are known to perform JLOCKs and cannot be used over a virtual terminal connection:

- VCRSAV, without the /T option.
- STRSAV.
- Any program during its PIC installation procedure.

The AMOS 2.0 BACKUP program **will** work correctly over a virtual terminal connection (except, of course, for video backup without /T).

Some programs may not work because of the difference in operating environments between a normal direct connection and a remote connection. For example, if you were to execute the BAUD rate command as a virtual connection, the baud rate of your direct connection to your terminal would not change.

5.2 LOCATING SYSTEMS AVAILABLE FOR CONNECTION

The FNDVT program can be used to locate remote systems which can be connected to by virtual terminal. Simply enter the FNDVT command at AMOS command level:

FNDVT RETURN

After a brief pause, you will see a display of the available systems, including their ersatz device name or cpuID, their symbolic name, the number of active jobs currently on the system, and the uptime of the system. For example:

```
Systems available for remote connection:

ACCNT: Accounting         126 active jobs, uptime 342:25:57
SALES: Sales Computer      15 active jobs, uptime 24:46:21
INVEN: Inventory System     7 active jobs, uptime 97:12:56
```

FNDVT will not find and display systems that must be accessed through gateways.

CHAPTER 6

MISCELLANEOUS AlphaNET COMMANDS

In this chapter, we discuss:

- How to Send Messages to Remote Computers
- How to Get Status Information About a Remote Computer
- How to Measure the Speed of a Network Connection

6.1 SENDING MESSAGES TO OTHER SYSTEMS

To send a message to someone on another computer, use the normal SEND command, but add a cpuID (or ersatz device name). The format is:

```
SEND cpuID-jobname MESSAGE
```

For example:

```
SEND 16842758-JESICA WHAT DID YOU WANT? RETURN
SEND 16842760-ROGER NEED TO TALK TO YOU ABOUT EDDIE RETURN
```

If you use ersatz device names in SEND, do not put a space after the colon.

What do you do when you want to send a message to someone over the network, but your not sure what their job name is? Use the RJOB command, this command will display the job names for the each user on the remote system. RJOB is described in this Chapter in Section 6.2.2.

6.2 GETTING STATUS INFORMATION ON REMOTE SYSTEMS

When using a network of several computers, you will often want to determine the status of a remote system, perhaps before attempting some special operation. Two programs, RSYS and RJOB, are provided to display such status information.

6.2.1 RSYS - Remote System Status

The RSYS program displays a brief status message on a specified system. It shows the number of active jobs on the system, and the total system uptime. This command is a quick way of determining if a specific system is up and available. To use RSYS, follow it with a cpuID or valid ersatz device name:

```
RSYS 16842759- RETURN
RSYS ARCHIV: RETURN
```

RSYS will respond with a message such as: `126 active jobs, uptime 628:37:41`. If the specified remote system is not available (it is not up or is not on the network), you will see a message to that effect.

6.2.2 RJOB - Remote Job Status

The RJOB program displays a brief list of active jobs on a remote system. It displays the job and terminal name, the user name, and the program being used. To use RJOB, follow it with a cpuID or valid ersatz device name:

```
RJOB 436842359- RETURN
RJOB ACCNT: RETURN
```

RJOB will respond with a display such as:

```
ACCT    JOB1    Accounting          DIR
TASK    MANAGR  System Service      TSKIDL
NETSER  NETSER  Network Service     NETSER
VTSER   VTSER   Network Service     VTSER
```

If the specified remote system is not available (it is not up or is not on the network), you will see a message to that effect.

6.3 MEASURING NETWORK CONNECTION SPEED

When using a collection of networks with gateways interconnecting them, it is sometimes useful to find out the relative speed of different connections. The NETSPD program measures and reports on the actual round-trip time needed to send a packet of information to a specified system. NETSPD has one option, NETSPD/L.

NETSPD displays the minimum, maximum, and average round-trip times, based on a sample of ten separate packets. The time displayed is expressed in milliseconds.

NETSPD will display a message such as: `Measured round trip packet transit time: Min = 43 ms, Max = 45 ms, Avg = 43.8 ms`. If the specified remote system is not available (not up or not on the network), it will tell you so.

To use NETSPD, follow it with a cpuID or valid ersatz device name:

NETSPD 16842759- RETURN
NETSPD ARCHIV: RETURN

NETSPD/L is more accurate and comprehensive way of measuring your network performance. The /L option allows you to send a 900 byte data packet of un-compressable data. You can also specify the number of packets you wish to send. If you do not specify the number of packets to be sent, the program will send the defaults quantity (10 packets).

To use NETSPD/L, follow it with a cpuID (or valid ersatz device name) and the number of packets you wish to send:

NETSPD/L 16842759- 100 RETURN
NETSPD/L ARCHIV: 100 RETURN

NETSPD has an error message that will warn the user when a packet is sent to a remote system and the packet echoed back is not the same packet. The error message reads:

```
?Sequence error. Expected n received x
```

Where n is the sequence number of the packet sent out and xis the sequence number of the packet received. This type of error gets generated when the network effectively gets out of sync and gets the packet order incorrect.

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